MODULE - 3: Model I. (General): $(M|M|1):(\infty|FCFS)$, and Model II. $(M|M|1):(N|FCFS)$

3.1 Model I (General) $(M|M|1):(\infty|FCFS)$

In this model, the arrival rate and service rate are assumed to be dependent on the system size n. The solution procedure of this model can be summarized in the following steps:

Step 1. **To obtain the system of steady state equations -** We take arrival rate $\lambda = \lambda_n$ and service rate $\mu = \mu_n$. Then, by the same arguments as in Model I, we have

$$\begin{aligned} P_n(t+\Delta t) &= P_n(t)[1-(\lambda_n+\mu_n)\Delta t] + P_{n-1}(t)\lambda_{n-1}\Delta t \\ &\quad + P_{n+1}(t)\mu_{n+1}\Delta t + o(\Delta t), \quad n>0; \\ \text{and} \quad P_0(t+\Delta t) &= P_0(t)[1-\lambda_0\Delta t] + P_1(t)\mu_1\Delta t + o(\Delta t), \quad n=0 \end{aligned}$$

Rearranging and dividing both sides by Δt, and then taking limit as $\Delta t \to 0$, we get

$$\begin{aligned} \frac{dP_n}{dt} &= -(\lambda_n+\mu_n)P_n(t) + \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t) \\ \text{and} \quad \frac{dP_0}{dt} &= -\lambda_0 P_0(t) + \mu_1 P_1(t). \end{aligned}$$

In the steady state, $P_n'(t) = 0$ and $P_0'(t) = 0$. Then we have

$$\begin{aligned} 0 &= -(\lambda_n+\mu_n)P_n + \lambda_{n-1}P_{n-1} + \mu_{n+1}P_{n+1}, \quad n>0 \\ \text{and} \quad 0 &= -\lambda_0 P_0 + \mu_1 P_1, \quad n=0. \end{aligned}$$

These equations constitute the *system of steady state difference equations* for this model.

Step 2. **To solve the system difference equations -**

We have

$$\begin{aligned}
P_1 &= \frac{\lambda_0}{\mu_1}P_0 \\
P_2 &= \frac{\lambda_1}{\mu_2}P_1 = \frac{\lambda_1\lambda_0}{\mu_2\mu_1}P_0 \\
P_3 &= \frac{\lambda_2}{\mu_3}P_2 = \frac{\lambda_2\lambda_1\lambda_0}{\mu_3\mu_2\mu_1}P_0 \\
\ldots & \quad \ldots\ldots \\
P_n &= \frac{\lambda_{n-1}\lambda_{n-2}\ldots\lambda_0}{\mu_n\mu_{n-1}\ldots\mu_1}P_0 \quad \text{for } n \geq 1
\end{aligned}$$

Since $\sum_{n=0}^{\infty} P_n = 1$, we have

$$\begin{aligned}
& P_0 + P_1 + P_2 + \ldots\infty = 1 \\
\text{or,} \quad & P_0\left[1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_1\lambda_0}{\mu_2\mu_1} + \ldots\infty\right] = 1 \\
\text{or,} \quad & P_0 = 1/S,
\end{aligned}$$

where $S = 1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_1\lambda_0}{\mu_2\mu_1} + \ldots\infty$.

Particular cases:

Case (i): $\lambda_{\mathbf{n}} = \lambda,\ \mu_{\mathbf{n}} = \mu$

In this case, the series S becomes

$$S = 1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \ldots\infty = \frac{1}{1 - \lambda/\mu}$$

Then

$$P_0 = \frac{1}{S} = 1 - \frac{\lambda}{\mu} \text{ and } P_n = \left(\frac{\lambda}{\mu}\right)^n\left(1 - \frac{\lambda}{\mu}\right)$$

which are the results obtained in Model I.

Case (ii): $\lambda_{\mathbf{n}} = \frac{\lambda}{\mathbf{n+1}}, \mu_{\mathbf{n}} = \mu$

In this case, the series S becomes

$$S = 1 + \frac{\lambda}{\mu} + \frac{\lambda^2}{2\mu^2} + \frac{\lambda^3}{2.3\mu^3} + \ldots\infty = 1 + \rho + \frac{\rho^2}{2!} + \frac{\rho^3}{3!} + \ldots\infty = e^{\rho}$$

e-Pathshala A Gateway to All Post Graduate Courses

Then we have

$$
\begin{aligned}
P_0 &= 1/S = e^{-\rho} \\
P_1 &= \frac{\lambda}{\mu} P_0 = \rho e^{-\rho} \\
P_2 &= \frac{\lambda^2}{2\mu^2} P_0 = \frac{\rho^2 e^{-\rho}}{2!} \\
\ldots & \quad \ldots\ldots \\
P_n &= \frac{\lambda^n}{n!\mu^n} P_0 = \frac{\rho^n e^{-\rho}}{n!} \quad \text{for all} \quad n = 0, 1, \ldots, \infty
\end{aligned}
$$

It is observed that P_n follows Poisson distribution, where $\lambda/\mu = \rho$, a constant. The value of ρ may be greater than or less than one but finite.

Case (iii): $\lambda_{\mathbf{n}} = \lambda,\ \mu_{\mathbf{n}} = \mathbf{n}\mu$

In this case, the arrival rate is independent of n, but the service rate μ_n increases as n increases. Here, we assume that there are infinite number of service stations. The word 'infinite' means that the service stations are available to each arrival. But, it does not mean that all the infinite service stations will remain busy every time.

In this case, the series S becomes

$$
S = 1 + \frac{\lambda}{\mu} + \frac{\lambda^2}{2\mu^2} + \frac{\lambda^3}{2.3\mu^3} + \ldots\infty = 1 + \rho + \frac{\rho^2}{2!} + \frac{\rho^3}{3!} + \ldots\infty = e^{\rho}
$$

Then $P_0 = e^{-\rho}$ and $P_n = \frac{e^{-\rho}\rho^n}{n!}$ which again follows Poisson distribution.

Example 3.1: Problems arrive at a computing centre in a Poisson fashion at an average rate 5 per day. The rules of the computing centre are that any man waiting to get his problem solved must aid the man whose problem is being served. If the time to solve a problem with one man has an exponential distribution with mean time of 1/3 day, and if the average solving time is inversely proportional to the number of people working on the problem, approximate the expected time in the centre for person entering the line.

Solution: Here $\lambda = 5$ problems/day, $\mu = 3$ problems/day. Then, the expected number

of persons working at any specified instant is

$$\begin{aligned} L_s &= \sum_{n=0}^{\infty} nP_n, \text{ where } P_n = \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n e^{-\lambda/\mu} \\ &= \sum_{n=0}^{\infty} n\frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n e^{-\lambda/\mu} = e^{-\lambda/\mu}\sum_{n=0}^{\infty}\frac{n}{n!}\left(\frac{\lambda}{\mu}\right)^n \\ &= e^{-\lambda/\mu}\cdot\frac{\lambda}{\mu}e^{\lambda/\mu} = \lambda/\mu = 5/3 \text{ persons.} \end{aligned}$$

Now, the average solving time which is inversly proportional to the number of people working on the problem is (1/5) day/problem.
Therefore, the expected time for a person entering the line is $(1/5)\times L_s$ days $= (1/5)\times(5/3)\times 24$ hours $= 8$ hours.

Example 3.2: A shipping company has a single unloading berth with ships arriving in Poisson fashion at an average rate of 3 per day. The unloading time distribution for a ship with the unloading crews is found to be exponential with average unloading time 1/2 day. The company has a large labour supply without regular working hours and to avoid long waiting lines, the company has a policy of using as many unloading crews as there are ships waiting in line or being unloaded.
Find (i) the average number of unloading crews working at any time and (ii) the probability that more than 4 crews will be needed.

Solution: Here $\lambda = 3$ ships/day, $\mu = 2$ ships/day. Then, the average number of unloading crews working at any specified instant is

$$L_s = \sum_{n=0}^{\infty} nP_n = \sum_{n=0}^{\infty} n\frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n e^{-\lambda/\mu} = \lambda/\mu = 3/2 \text{ crews.}$$

The probability that more than 4 crews will be needed is the same as probability that there are at least 5 ships in the system at any specified time which is given by

$$\begin{aligned} \sum_{n=5}^{\infty} P_n &= \sum_{n=0}^{\infty} P_n - \sum_{n=0}^{4} P_n, \text{ where } P_n = \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n e^{-\lambda/\mu} \\ &= 1 - [e^{-\lambda/\mu} + \left(\frac{\lambda}{\mu}\right)\frac{e^{-\lambda/\mu}}{1!} + \left(\frac{\lambda}{\mu}\right)^2\frac{e^{-\lambda/\mu}}{2!} + \ldots + \left(\frac{\lambda}{\mu}\right)^4\frac{e^{-\lambda/\mu}}{4!}] = 0.019. \end{aligned}$$

3.2 Model II. $(M|M|1):(N|FCFS)$

This model is based on all assumptions of Model I, except a limit on the capacity of the system to accommodate only N customers. This implies that once the line reaches

its maximum length of N customers, no additional customers will be allowed to enter into the system. Therefore, the difference equations of Model I are valid for this model as long as $n < N$. The additional difference equation for $n = N$ is

$$P_N(t+\Delta t) = P_N(t)[1-\mu\Delta t] + P_{N-1}(t).[\lambda\Delta t][1-\mu\Delta t] + o(\Delta t)$$

The corresponding differential-difference equation can be obtained as

$$\frac{dP_N}{dt} = -\mu P_N(t) + \lambda P_{N-1}(t)$$

from which the resultant steady-state difference equation is

$$0 = -\mu P_N + \lambda P_{N-1}$$

The complete set of steady-state difference equations for this model, therefore, can be written as

$$\begin{aligned} \mu P_1 &= \lambda P_0, \quad n = 0 \\ \mu P_{n+1} &= (\lambda+\mu)P_n - \lambda P_{N-1}, \quad 1 \le n \le N-1 \\ \text{and} \qquad \mu P_N &= \lambda P_{N-1}, \quad n = N. \end{aligned}$$

In this case, the service rate (μ) should not exceed the arrival rate (λ) in order to obtain the steady-state conditions. From the first two difference equations, the probability of n ($n = 0, 1, 2, ..., N-1$) customers in the system is obtained as

$$P_n = (\lambda/\mu)^n P_0; \quad n \le N-1.$$

Also, we see that for this value of P_n, the third difference equation holds for $n = N$. Therefore, we have

$$P_n = (\lambda/\mu)^n P_0; \quad n \le N.$$

To obtain the value of P_0, we use the fact that

$$\begin{aligned} 1 &= \sum_{n=0}^{N} P_n = \sum_{n=0}^{N} (\lambda/\mu)^n P_0 \\ &= P_0 \sum_{n=0}^{N} \rho^n = P_0 \left[\frac{1-\rho^{N+1}}{1-\rho} \right] \end{aligned}$$

Consequently, we have

$$\begin{aligned} P_0 &= \begin{cases} \frac{1-\rho}{1-\rho^{N+1}}; & \rho \ne 1 \\ \frac{1}{N+1} & \rho = 1 \end{cases} \\ P_n &= \begin{cases} \left(\frac{1-\rho}{1-\rho^{N+1}}\right)\rho^n; & \rho \ne 1 \\ \frac{1}{N+1} & \rho = 1 \end{cases} \end{aligned}$$

Note: *If $N \to \infty$, then the steady state solution is*

$$P_n = (1-\rho)\rho^n; \quad n < \infty$$

We see that this result is same as that of Model I.

- **Characteristics of Model II**

(i) Average number of customers in the system is given by

$$\begin{aligned} L_s &= \sum_{n=0}^{N} nP_n = P_0 \sum_{n=0}^{N} n\rho^n = P_0\rho \sum_{n=0}^{N} \frac{d\rho^n}{d\rho} \\ &= P_0\rho \frac{d}{d\rho} \sum_{n=0}^{N} \rho^n = P_0\rho \frac{d}{d\rho}\left[\frac{1-\rho^{N+1}}{1-\rho}\right] \\ &= P_0 \frac{\rho[1-(N+1)\rho^N + N\rho^{N+1}]}{(1-\rho)^2} \\ &= \frac{\rho[1-(N+1)\rho^N + N\rho^{N+1}]}{(1-\rho)(1-\rho^{N+1})} \end{aligned}$$

(ii) $L_q = L_s - \lambda/\mu$

(iii) $W_s = L_s/\lambda$

(iv) $W_q = W_s - 1/\mu = L_q/\lambda$

Example 3.3: Consider a single server queuing system with Poisson input and exponential service time. Suppose that the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hour and the maximum permissible calling units in the system is 2. Derive the steady state probability distribution of the number of calling units in the system, and then calculate the expected number in the system.

Solution: Given that $\lambda = 3$ units per hour; $\mu = 4$ units per hour, and $N = 2$. Then traffic intensity $\rho = \lambda/\mu = 3/4 = 0.75$. The steady state probability distribution of the number of n customers (calling units) in the system is

$$\begin{aligned} P_n &= \frac{(1-\rho)\rho^n}{1-\rho^{N+1}} = \frac{(1-0.75)(0.75)^n}{1-(0.75)^{2+1}} = (0.43)(0.75)^n; \quad \rho \neq 1 \\ \text{and} \quad P_0 &= \frac{(1-\rho)}{1-\rho^{N+1}} = \frac{(1-0.75)}{1-(0.75)^{2+1}} = \frac{0.25}{1-(0.75)^3} = 0.431 \end{aligned}$$

The expected number of calling units in the system is given by

$$\begin{aligned} L_s &= \sum_{n=1}^{N} nP_n = \sum_{n=1}^{2} n(0.43)(0.75)^n \\ &= 0.43 \sum_{n=1}^{2} n(0.75)^n = 0.43\{(0.75) + 2(0.75)^2\} = 0.81. \end{aligned}$$

e-Pathshala पाठशाला A Gateway to All Post Graduate Courses

Example 3.4: Assume that the goods trains are coming in a yard at the rate of 30 trains per day and suppose that the inter-arrival times follow an exponential distribution. The service time for each train is assumed to be exponential with an average of 36 minutes. If the yard can admit 9 trains at a time (there being 10 lines, one of which is reserved for shunting purposes), calculate the probability that the yard is empty and find the average queue length.

Solution: Given that $\lambda = \frac{30}{60\times 24} = \frac{1}{48}$ train per minute and $\mu = \frac{1}{16}$ train per minute.
Therefore, $\rho = \lambda/\mu = 36/48 = 0.75$.
The probability that the yard is empty is given by

$$P_0 = \frac{1-\rho}{1-\rho^{N+1}} = \frac{1-0.75}{1-(0.75)^{10}} = 0.28.$$

Average queue length is given by

$$\begin{aligned} L_q &= \frac{\rho^2[1-N\rho^{N-1}+(N-1)\rho^N]}{(1-\rho)(1-\rho^{N+1})} \\ &= \frac{(0.75)^2[1-9(0.75)^8+8(0.75)^9]}{0.25(1-0.75^{10})} = 1.55 \end{aligned}$$

epathshala पाठशाला A Gateway to All Post Graduate Courses